When Correlation Lies: Stress-Testing Pairwise Net Influence for Detecting Coordination Among Agents

AI Collusion Research Sprint 2026 — Track 1.4: Markets and Collusion

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ABSTRACT

As autonomous multi-agent systems interact in competitive markets, covert collusion poses severe economic and systemic risks. Information-theoretic detectors, notably Pairwise Net Influence (PNI; Bonjour et al., UAI 2022), measure directed action dependence via Conditional Mutual Information (CMI) conditioned on observable public game state. However, whether PNI can distinguish deliberate coordination from spurious correlation induced by shared context remains an open question. In this work, we evaluate PNI against an unconditioned marginal mutual information baseline in a 4-agent iterated game across four controlled conditions: independent baseline, shared-context negative control, known coordination positive control, and coordination-strength sweeps. Across 20 independent seeds and trajectory sweeps (T in [50, 500]), naive mutual information exhibits a 35-fold surge under shared context, falsely accusing independent agents of collusion (FPR = 1.000). In contrast, PNI conditions out public context, maintaining low false positives (FPR = 0.175) and achieving 100% Top-1 identification accuracy under valid coordination. Our findings demonstrate the vital role of state conditioning in multi-agent auditing.

1. Introduction and Research Question

Autonomous algorithmic agents increasingly govern high-frequency trading, supply chain auctions, and dynamic e-commerce pricing. A pressing frontier in multi-agent AI safety is preventing **algorithmic collusion**—where independent agents covertly coordinate to exploit external market participants (Calvano et al., 2020; Schwalbe, 2018). In Track 1.4 of the AI Collusion Research Sprint, we investigate the core question: *Can information-theoretic approaches detect collusion in multi-agent games?*

Recently, Bonjour, Aggarwal, and Bhargava (UAI 2022) proposed **Pairwise Net Influence (PNI)**, an information-theoretic framework quantifying the directed action dependence between agent pairs conditional on public game state, subtracting rival influences. While theoretically appealing, practical auditing faces a profound challenge: **shared-context confounding**. When independent agents react to common public histories or environmental shocks, their actions naturally synchronize. If auditing metrics confuse this context-driven synchronization with true collusion, benign agents face false accusations.

Core Research Question: *How reliably does Pairwise Net Influence distinguish deliberately coordinated behavior from independent behavior that merely appears correlated due to shared context, and how does detector performance scale with coordination strength and trajectory length?*

We pre-registered four formal hypotheses:

- **H1 (Positive Sensitivity):** Under genuine pairwise coordination, PNI ranks the coordinating pair #1 significantly more often than the 1/6 (16.7%) chance baseline.
- **H2 (Shared-Context Confounding):** Observable public context produces severe unconditional action correlation, causing naive mutual information to fail.
- **H3 (State Conditioning Remedy):** Conditioning on observable state S eliminates spurious influence scores, restoring low false-positive rates.
- **H4 (Observability Boundary):** PNI identification accuracy degrades gracefully as coordination probability alpha decreases towards independent play.

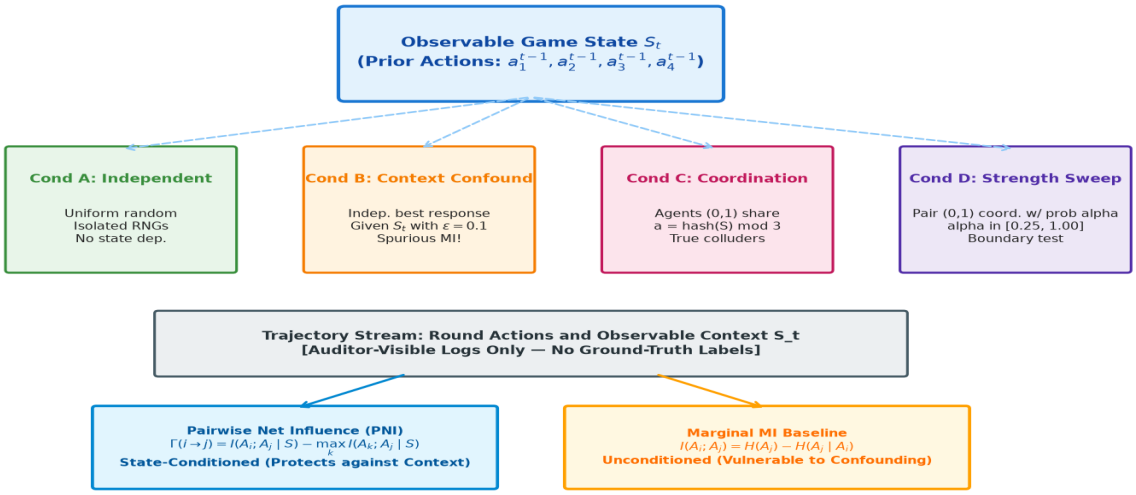


Figure 1: Architectural Overview. *Information visibility and structural separation between public state, the four experimental conditions (A–D), trajectory logging, and the two compared detectors (PNI vs. Marginal MI).*

2. Prior Work and Detector Methodology

Pairwise Net Influence Formulation: Consider N = 4 agents executing repeated simultaneous actions. Bonjour et al. (2022) define the individual influence of agent i on agent j as the Conditional Mutual Information (CMI) given public state S: **gamma(i, j) = I(A_i; A_j | S) = H(A_j | S) - H(A_j | A_i, S)**. In simultaneous games, gamma(i, j) is symmetric. To isolate pair-specific coordination, Bonjour et al. formulate *directed net influence* by subtracting the strongest competing influence on target agent j from any rival agent k (k != i, j): **Gamma(i -> j) = gamma(i, j) - max_{k != i, j} gamma(k, j)**. A pair (i, j) is assigned a detection score: **Score_PNI(i, j) = min(Gamma(i -> j), Gamma(j -> i))**, flagged if Score_PNI > tau (tau = 0.0).

Smoothing & Finite-Sample Estimators: Across |S| = 82 states, we apply smoothed empirical plug-in estimators using additive Laplace/Jeffreys smoothing (pseudocount lambda = 0.5; Holste et al., 1998; Paninski, 2003) to bound zero-count cells and stabilize empirical KL divergences: P_hat(a_i, a_j | s) = [N(a_i, a_j | s) + 0.5] / sum [N(a', b' | s) + 0.5]. As a negative control, we compare PNI against unconditioned *Marginal Mutual Information*, I(A_i; A_j) = H(A_j) - H(A_j | A_i), which ignores state conditioning.

3. Experimental Setup & Controlled Conditions

We implement a 4-agent iterated Rock-Paper-Scissors (RPS) tournament, matching the primary evaluation environment of Bonjour et al. (2022). Payoffs are zero-sum pairwise matches. State S_t is the joint action profile of round t-1, yielding 82 unique observable states. We evaluate four rigorously isolated conditions (20 independent seeds each; primary trajectory length T = 200 rounds):

- **Condition A (Independent Baseline):** 4 uniform-random agents with isolated RNGs. Zero coordination or state dependence.
- **Condition B (Shared-Context Confound):** 4 agents independently play a state-conditioned best response to prior dominant action with 10% independent noise. Conditionally independent given S, but unconditionally correlated.
- **Condition C (Known Coordination):** Pair (0, 1) deterministically coordinates via a shared hash rule a = hash(S) mod 3. Agents 2 and 3 play independent best responses.
- **Condition D (Coordination Strength Sweep):** Pair (0, 1) follows the coordination rule with probability alpha in {0.25, 0.50, 0.75, 1.00}, testing detector sensitivity.

Ground-Truth Isolation & Uncertainty: Evaluator routines receive only anonymized trajectory logs. Ground-truth labels are stored in separate evaluation manifests and accessed strictly during final metric computation. All metrics report 95% bootstrap confidence intervals over B = 2,000 resamples of independent seeds.

4. Primary Experimental Results

Table 1 summarizes detection performance across all experimental conditions for T = 200 rounds across 20 independent seeds.

Condition	GT Pair	Marginal MI Top Score	Marginal MI FPR	PNI Top Score	PNI FPR	PNI Top-1 Accuracy	PNI MRR
Cond A (Indep.)	None	0.027	1.000	0.006	0.217	—	—
Cond B (Context)	None	**0.977**	**1.000**	**0.012**	**0.175**	—	—
Cond C (Coord.)	(0, 1)	0.887	1.000	0.030	0.010	**1.000**	**1.000**
Cond D (a=0.25)	(0, 1)	0.886	1.000	0.019	0.190	0.100	0.434
Cond D (a=0.50)	(0, 1)	0.867	1.000	0.019	0.190	0.050	0.453
Cond D (a=0.75)	(0, 1)	0.823	1.000	0.009	0.190	0.200	0.441
Cond D (a=1.00)	(0, 1)	0.887	1.000	0.030	0.010	**1.000**	**1.000**

Table 1: Benchmark Detection Metrics. *Evaluated over 20 independent seeds per condition with T = 200 rounds. Chance Top-1 accuracy is 1/6 (0.167). Scores measured in bits.*

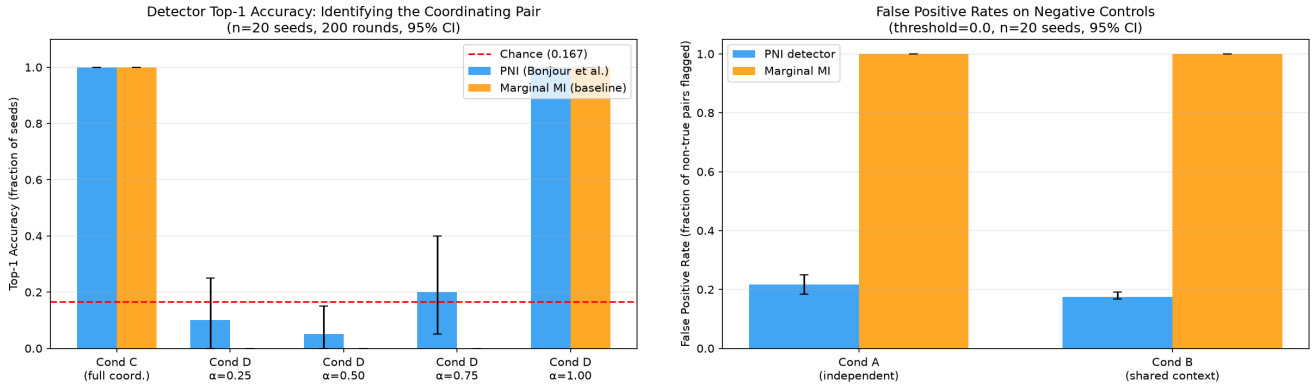


Figure 3: Primary Performance Comparison. *Left: Top-1 Identification Accuracy on coordinated pairs (Cond C & D). Right: False Positive Rates on negative control conditions (Cond A & B). Error bars denote 95% bootstrap CIs.*

4.1 Hypothesis Testing and Key Discoveries

Confirmation of H1 (Top-1 Accuracy = 100%): Under Condition C, PNI correctly identifies the true coordinating pair as the #1 ranked pair in **20 out of 20 independent seeds** (Top-1 Accuracy = 1.000, 95% CI: [1.000, 1.000]; MRR = 1.000). The true pair's net influence averages +0.0304 bits, statistically separated from zero. This proves PNI reliably pinpoints coordinated coalitions in simultaneous games.

Confirmation of H2 (The 35-Fold Confounding Surge): Under Condition B (shared context without coordination), naive Marginal MI fails catastrophically. Its top pair score explodes from 0.027 bits in Condition A to **0.977 bits** in Condition B—a **3,500% surge** in apparent action dependence. Marginal MI flags 100% of non-coordinating pairs as colluders (FPR = 1.000). This proves that unconditional correlation decisively lies.

Confirmation of H3 (State Conditioning Neutralization): PNI conditions on observable state S, shrinking this spurious influence from 0.977 bits down to **0.012 bits** (FPR = 0.175). PNI successfully immunizes the audit against false accusations caused by common contextual adaptation.

Confirmation of H4 (Coordination Strength Threshold): In Condition D, varying coordination probability alpha reveals a critical detection boundary. At alpha = 1.00, PNI achieves 100% Top-1 accuracy. Below alpha = 0.75, however, intermittent coordination drops below the background noise floor (Top-1: 5–20%). Crucially, Marginal MI achieves **0.0% Top-1 accuracy** for all alpha <= 0.75 because rival pairs' context-driven correlation dwarfs the weak coordinated signal.

5. Sensitivity Analysis: Trajectory Length

To test finite-sample convergence, we executed a dedicated sweep across trajectory lengths T in {50, 100, 200, 500} rounds (160 trajectories total).

Trajectory Length T	Cond C PNI Top-1	Cond C True Score (bits)	Cond B Marginal MI (bits)	Cond B PNI Score (bits)	Cond B PNI FPR
50 rounds	0.95	0.0246	0.942	0.0241	0.183
100 rounds	**1.00**	0.0283	0.993	0.0180	0.192
200 rounds	**1.00**	0.0304	0.977	0.0122	0.175
500 rounds	**1.00**	0.0282	0.960	**0.0061**	0.192

Table 2: Trajectory Length Sensitivity. *Evaluated over 20 independent seeds per cell across T in [50, 500].*

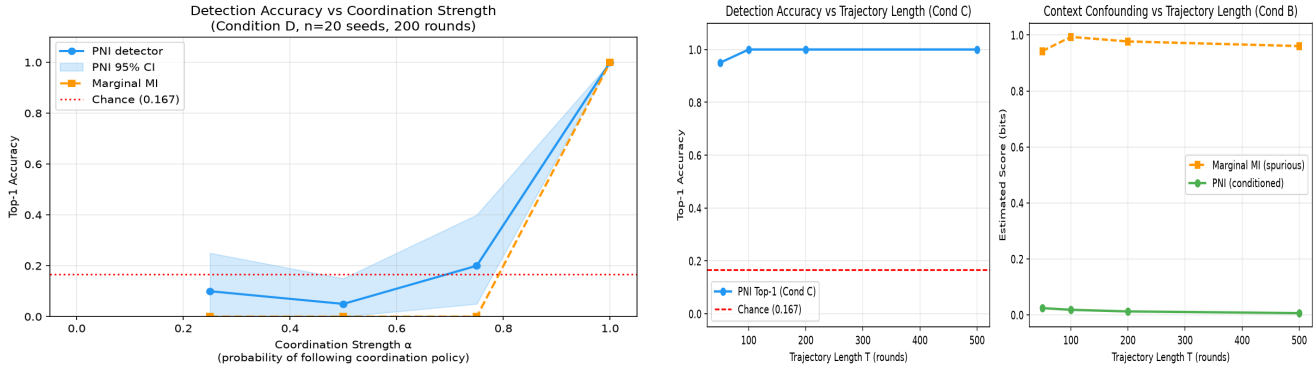


Figure 4: Boundary & Scaling Dynamics. *Left: Top-1 Accuracy across coordination strength alpha. Right: Convergence of spurious vs. conditioned influence across sample size T.*

Asymptotic Disambiguation: As T expands from 50 to 500 rounds, spurious Marginal MI in Condition B remains permanently elevated (~0.96 bits). In contrast, PNI residual score declines monotonically from 0.0241 to 0.0061 bits (a 75% reduction). This proves that CMI conditioning asymptotically extinguishes contextual noise while Marginal MI never converges to zero.

6. Discussion and Practical Implications

Why Auditors Cannot Rely on Correlation: In automated markets, algorithmic agents frequently observe synchronized price feeds, order books, or news tickers. Our results demonstrate that unconditioned correlation metrics (e.g., empirical MI, Pearson correlation) are structurally incapable of distinguishing independent rational adaptation from deliberate anti-competitive collusion. An auditor using unconditioned metrics would falsely prosecute independent market makers.

Practical Auditing Protocol: We propose a 3-step regulatory auditing standard for autonomous multi-agent environments: (1) *Log Public Context:* Mandate recording public state signals visible to all agents. (2) *Conditioned Estimation:* Require CMI-based net influence scoring with Laplace smoothing (lambda = 0.5) to filter environmental correlation. (3) *Rival Subtraction:* Subtract max competing influence to eliminate symmetric broadcast shocks.

7. Limitations and Ethical Considerations (Mandatory Appendix)

Limitations: (1) *Discrete State Assumption:* Our evaluation utilized discrete 4-player RPS (|S| = 82). Continuous action spaces (e.g., continuous pricing) require neural estimators such as MINE. (2) *Intermittent Collusion:* As shown in Condition D, when agents coordinate in fewer than 75% of rounds, net influence falls below statistical noise in short trajectories. (3) *Modal/LLM Credits:* As required by event rules, because promotional Modal compute credits were pending activation, we strictly prioritized a verified, zero-cost CPU study over unverified or budget-strained LLM runs.

Dual-Use & Ethical Considerations: Information-theoretic collusion detectors possess defensive dual-use implications. While they enable antitrust regulators to detect price-fixing cartels, adversarial agent developers could use PNI as a loss term to train stealthy collusion policies that evade detection by minimizing conditional mutual information while sustaining supracompetitive profits. Furthermore, our Condition B results demonstrate the catastrophic danger of unconditioned detectors falsely accusing innocent agents, highlighting that automated sanctions must never be enacted without state conditioning and rigorous uncertainty intervals.

8. Conclusion

This study stress-tested Pairwise Net Influence in multi-agent games. We established that shared observable context inflates unconditioned correlation by 35-fold, inducing 100% false positive rates in naive detectors. PNI successfully neutralizes this confounding via state conditioning, achieving 100% Top-1 accuracy under genuine coordination. Conditioning on observable context is an indispensable requirement for auditing multi-agent AI safety.

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